

Application Guide

VACON® 100 FLOW



Contents

1	Introduction	11
1.1	Purpose of this Application Guide	11
1.2	Manual and Software Version	11
1.3	Additional Resources	11
1.4	Parameter Table Reading Guide	11
1.5	Functions of the AC Drive	12
1.6	Start-up Quick Guide	13
1.6.1	Start-Up Wizard	13
2	User Interfaces	14
2.1	Description of the Control Panel	14
2.1.1	Control Panel and the Keypad	14
2.1.2	Displays of the Control Panel	14
2.2	Menu Structure	15
2.3	Graphical Display	18
2.3.1	Editing the Text Values	18
2.3.2	Editing the Numerical Values	18
2.3.3	Selecting More than 1 Value	19
2.3.4	Resetting a Fault	19
2.3.5	[FUNCT] Button	19
2.3.5.1	Different Control Places	19
2.3.5.2	Changing the Control Place	20
2.3.5.3	Going into the Control Page	20
2.3.5.4	Changing the Rotation Direction	21
2.3.5.5	Using the Quick Edit Function	21
2.3.6	Copying the Parameters of an AC Drive	22
2.3.7	Comparing the Parameters	22
2.3.8	Reading Help Texts	23
2.3.9	Using the Favourites Menu	24
2.4	Text Display	24
2.4.1	Editing the Text Values	24
2.4.2	Editing the Numerical Values	24
2.4.3	Resetting a Fault	19
2.4.4	[FUNCT] Button	25
2.4.4.1	Different Control Places	19
2.4.4.2	Changing the Control Place	25
2.4.4.3	Going into the Control Page	26

2.4.4.4	Changing the Rotation Direction	26
2.4.4.5	Using the Quick Edit Function	21
2.5	VACON® Live PC Tool	27
3	Applications	28
3.1	Using the Applications	28
3.2	Standard and HVAC Applications	28
3.3	PID Control Application	32
3.4	Multi-pump (Single Drive) Application	36
3.5	Multi-pump (Multidrive) Application	43
4	Wizards	67
4.1	Application Wizards	67
4.1.1	Standard Application Wizard	67
4.1.2	HVAC Application Wizard	67
4.1.3	PID Control Application Wizard	68
4.1.4	Multi-pump (Single Drive) Application Wizard	69
4.1.5	Multi-pump (Multidrive) Application Wizard	70
4.2	Function Wizards	72
4.2.1	Wizards in Menu 1.1	72
4.2.2	Fire Mode Wizard	72
5	Monitor Menu	73
5.1	Uses of the Monitor Menu	73
5.2	Multimonitor Monitoring	73
5.2.1	Changing the Items to Monitor	73
5.3	Trend Curve Monitoring	73
5.3.1	Uses of the Trend Curve	73
5.3.2	Changing the Values in Trend Curve Monitoring	73
5.3.3	Stopping the Progression of the Curve	74
5.3.4	Trend Curve Parameters	75
5.4	Basic Monitoring	75
5.4.1	Uses of the Basic Monitoring	75
5.4.2	Basic Monitoring Parameters	75
5.5	I/O Monitoring	76
5.6	Temperature Inputs	76
5.7	Extras and Advanced	77
5.8	Timer Functions Monitoring	79
5.9	PID Controller Monitoring	79

5.10	External PID Controller Monitoring	80
5.11	Multi-pump Monitoring	80
5.12	Monitoring of Maintenance Counters	81
5.13	Fieldbus Process Data Monitoring	81
5.14	Drive Customizer Monitoring	82
5.15	Condition-based Monitoring	83
6	Parameters Menu	85
6.1	Group 3.1: Motor Settings	85
6.2	Group 3.2: Start/Stop Setup	87
6.3	Group 3.3: References	88
6.4	Group 3.4: Ramps and Brakes Setup	91
6.5	Group 3.5: I/O Configuration	92
6.6	Group 3.6: Fieldbus Data	99
6.7	Group 3.7: Prohibit Frequencies	101
6.8	Group 3.8 Supervisions	101
6.9	Group 3.9: Protections	102
6.10	Group 3.10: Automatic Reset	105
6.11	Group 3.11: Application Settings	106
6.12	Group 3.12: Timer Functions	107
6.13	Group 3.13: PID Controller	109
6.14	Group 3.14: External PID Controller	117
6.15	Group 3.15: Multi-pump	120
6.16	Group 3.16: Maintenance Counters	122
6.17	Group 3.17: Fire Mode	123
6.18	Group 3.18: Motor Preheat Parameters	124
6.19	Group 3.19: Drive Customizer	124
6.20	Group 3.21: Pump Control	124
6.21	Group 3.23: Advanced Harmonic Filter	127
6.22	Group 3.24: Condition-based Monitoring	127
7	Diagnostics Menu	132
7.1	Faults	132
7.1.1	Uses of the Active Faults Submenu	132
7.1.2	Reset Faults	132
7.1.2.1	Resetting Faults with the [BACK/RESET] Button	132
7.1.2.2	Resetting Faults with a Parameter on the Graphical Display	132
7.1.2.3	Resetting Faults with a Parameter on the Text Display	132
7.1.3	Uses of the Fault History	133

7.1.3.1	Examining the Fault History on the Graphical Display	133
7.1.3.2	Examining the Fault History on the Text Display	133
7.2	Counters	134
7.2.1	Using the Counters	134
7.2.2	Total Counters	134
7.2.3	Trip Counters	135
7.3	Software Info	136
7.4	Drive Info	136
7.5	Unit Status	136
8	I/O and Hardware Menu	137
8.1	Basic I/O	137
8.2	Option Board Slots	137
8.3	Real-time Clock	138
8.4	Simple Network Time Protocol (SNTP)	138
8.5	Power Unit Settings	139
8.6	Keypad	139
8.7	Fieldbus	140
8.8	Fieldbus General	142
8.9	STO	143
8.10	Fieldbus Redundancy	143
9	User Settings, Favourites, and User Level Menus	144
9.1	User Settings	144
9.1.1	User Settings Parameters	144
9.1.2	Parameter Back-up Parameters	144
9.1.3	Parameter Compare Parameters	145
9.1.4	License Parameters	145
9.2	Favourites	145
9.2.1	Using the Favourites Menu	145
9.2.2	Adding an Item to the Favourites	145
9.2.3	Removing an Item from the Favourites	145
9.3	User Levels	146
9.3.1	Using the User Levels Menu	146
9.3.2	The User Level Parameters	146
9.3.3	Changing the Access Code of the User Levels	146
10	Monitoring Value Descriptions	148
10.1	Multimonitor	148

10.2	Basic	149
10.3	I/O	150
10.4	Temperature Inputs	151
10.5	Extras and Advanced	152
10.6	Timer Functions	153
10.7	PID Controller	154
10.8	External PID Controller	155
10.9	Multi-pump	155
10.10	Maintenance Counters	156
10.11	Fieldbus Data	156
10.12	Drive Customizer	159
10.13	Condition-based Monitoring	160
10.13.1	Baseline Data	160
10.13.2	Stator Winding Monitor	161
10.13.3	Vibration	161
10.13.4	Load	162

11 Parameter Descriptions 163

11.2	Trend Curve	163
11.3	Motor Settings	164
11.3.1	Motor Nameplate Parameters	164
11.3.2	Motor Control Parameters	164
11.3.3	Motor Limits	168
11.3.4	Open Loop Parameters	168
11.3.5	I/f Start Function	171
11.3.6	Torque Stabilator Function	172
11.4	Start/Stop Setup	173
11.4.1	Start and Stop Commands in Different Control Places	173
11.5	References	181
11.5.1	Frequency Reference	181
11.5.2	Preset Frequencies	182
11.5.3	Motor Potentiometer Parameters	185
11.5.4	Flushing Parameters	186
11.6	Ramps and Brakes Setup	186
11.6.1	Ramp 1	186
11.6.2	Ramp 2	187
11.6.3	Start Magnetizing	189
11.6.4	DC Brake	189

11.6.5	Flux Braking	189
11.7	I/O Configuration	190
11.7.1	Programming of Digital and Analog Inputs	190
11.7.1.1	General Information on the Programming	190
11.7.1.2	Programming of Digital Inputs	191
11.7.1.3	Programming of Digital Inputs in the Graphical Display	193
11.7.1.4	Programming of Digital Inputs in the Text Display	193
11.7.1.5	Programming of Analog Inputs	194
11.7.1.6	Programming of Analog Inputs in the Graphical Display	195
11.7.1.7	Programming of Analog Inputs in the Text Display	196
11.7.2	Descriptions of Signal Sources	196
11.7.3	Default Functions of Programmable Inputs	197
11.7.4	Digital Inputs	197
11.7.5	Analog Inputs	202
11.7.6	Digital Outputs	206
11.7.7	Analog Outputs	209
11.8	Fieldbus Data	212
11.9	Prohibit Frequencies	214
11.10	Supervisions	216
11.11	Protections	216
11.11.1	General	216
11.11.2	Motor Thermal Protections	218
11.11.3	Motor Stall Protection	221
11.11.4	Underload (Dry Pump) Protection	222
11.11.5	Quick Stop	224
11.11.6	AI Low Protection	225
11.12	Automatic Reset	226
11.13	Application Settings	229
11.14	Timer Functions	229
11.14.1	Introduction to Timer Functions	229
11.14.2	Time Channels, Intervals, and Timers	230
11.14.3	Editing an Interval	232
11.15	PID Controller	233
11.15.1	Basic Settings	233
11.15.2	Setpoints	235
11.15.3	Feedback	236
11.15.4	Feedforward	236
11.15.5	Sleep Function	237

11.15.6 Feedback Supervision	240
11.15.7 Pressure Loss Compensation	242
11.15.8 Soft Fill	243
11.15.9 Input Pressure Supervision	244
11.15.10 Sleep Function When No Demand Is Detected	246
11.15.11 Multi-setpoint	248
11.16 External PID Controller	249
11.17 Multi-pump Function	250
11.17.1 The Multi-pump Function	250
11.17.1.1 Commissioning Multi-pump (Multidrive)	250
11.17.2 System Configuration	251
11.17.3 Interlocks	256
11.17.4 Feedback Sensor Connection in a Multi-pump System	256
11.17.5 Overpressure Supervision	263
11.17.6 Pump Run-time Counters	264
11.17.7 Advanced Settings	265
11.18 Maintenance Counters	266
11.18.1 The Maintenance Counters	266
11.19 Fire Mode	267
11.19.1 The Fire Mode Function	267
11.20 Motor Preheat Function	270
11.21 Drive Customizer	271
11.22 Pump Control	271
11.22.1 Auto-Cleaning	271
11.22.2 Jockey Pump	273
11.22.3 Priming Pump	274
11.22.4 Anti-blocking Function	275
11.22.5 Frost Protection	275
11.22.6 Flow Switch	276
11.23 Advanced Harmonic Filter	277
11.24 Condition-based Monitoring	277
11.24.1 Introduction to Condition-based Monitoring	277
11.24.1.1 Stages of Condition-based Monitoring	278
11.24.6 Manual Run	279
11.24.7 Stator Winding Monitoring	280
11.24.8 Threshold Value	280
11.24.9 Warning and Alarm/Fault Stages	281
11.25 Default Values of Parameters in the Different Applications	286

12 Fault Tracing	292
12.1 Fault Types	292
12.2 Alarms and Warnings	292
12.3 Fault Codes	292
12.4 Total and Trip Counters	317
12.4.1 Operating Time Counter	317
12.4.2 Operating Time Trip Counter	318
12.4.3 Run Time Counter	318
12.4.4 Power on Time Counter	319
12.4.5 Energy Counter	319
12.4.6 Energy Trip Counter	320

1 Introduction

1.1 Purpose of this Application Guide

This Application Guide provides information for configuring the system, controlling the AC drive, accessing parameters, programming, and troubleshooting of the AC drive. It is intended for use by qualified personnel. Read and follow the instructions to use the drive safely and professionally. Pay particular attention to the safety instructions and general warnings that are provided in this manual and other documentation delivered with the drive.

1.2 Manual and Software Version

This manual is regularly reviewed and updated. All suggestions for improvement are welcome.

The original language of this manual is English.

Table 1: Manual and Software Version

Edition	Software version
DPD01083H	FW0159V024

1.3 Additional Resources

Other resources are available to understand advanced AC drive functions and operation.

- VACON® 100 Wall-mounted Drives Operating Guide
- VACON® 100 Enclosed Drives Installation Manual
- VACON® 100 IP00 Drive Modules Installation Manual
- VACON® 100 X Installation Manual
- Instructions for operation with option boards and other optional equipment.

Supplementary publications and manuals are available from Danfoss.

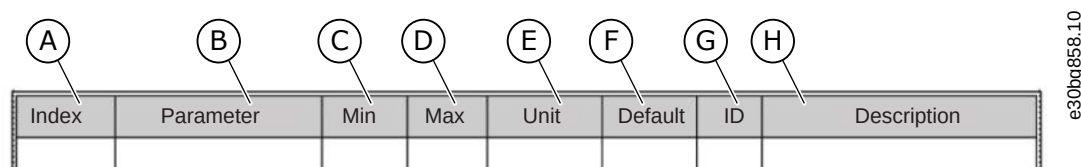
For US and Canadian markets:

NOTE! Download the English and French product guides with applicable safety, warning and caution information from <https://www.danfoss.com/en/service-and-support/>.

REMARQUE Vous pouvez télécharger les versions anglaise et française des guides produit contenant l'ensemble des informations de sécurité, avertissements et mises en garde applicables sur le site <https://www.danfoss.com/en/service-and-support/>.

1.4 Parameter Table Reading Guide

This manual includes a large quantity of parameter tables. These instructions tell you how to read the tables.



A	B	C	D	E	F	G	H
Index	Parameter	Min	Max	Unit	Default	ID	Description

Illustration 1: Parameter Table Reading Guide

A	The location of the parameter in the menu, that is, the parameter number.	E	The unit of the value of the parameter. The unit shows if it is available.
B	The name of the parameter.	F	The value that was set in the factory.
C	The minimum value of the parameter.	G	The ID number of the parameter.
D	The maximum value of the parameter.	H	A short description of the values of the parameter and/or its function.

1.5 Functions of the AC Drive

- Easy setup functions
 - Applications with preset parameters and I/O configuration for your process: Standard, HVAC, PID control, Multi-pump
 - Start-up wizards for fast and easy parameterization
- Navigation functions
 - The [FUNCT] button for an easy change between the local and the remote control place (I/O or fieldbus)
 - A control page to operate and monitor of the most important values quickly
- Advanced monitoring functions
 - Multimonitor
 - Trend curve
- Fieldbus and I/O control
- Motor control
 - Open loop control with frequency reference
 - Support for induction, permanent magnet, and synchronous reluctance motors
 - Identification run for setup of motor parameters
 - Flux and DC brake
- Start and stop logic configuration
- Reference selection and handling
 - 8 preset frequencies
 - Motor potentiometer functions
 - A flush function
 - 2 programmable ramp times
 - 2 supervisions
 - 3 ranges of prohibited frequencies
 - A forced stop
- PID control
 - An external PID controller is available. Use it, for example, to control a valve with the I/O of the AC drive.
 - A sleep mode function that automatically enables and disables the operation of the drive to save energy
 - A 2-zone PID controller with 2 different feedback signals: minimum and maximum control
 - 2 setpoint sources for the PID control. Selection to be made with a digital input.
 - A function for PID setpoint boost
 - A feedforward function to make the response to the process changes better
- Pump control
 - A multi-pump control for the single drive and multidrive systems
 - The multimaster and multifollower modes in the multidrive system
 - A multi-pump system that uses a real time clock to autochange the pumps
 - Priming pump control, jockey pump control, pump impeller auto-cleaning, pump input pressure supervision, and frost protection function
- Mechanical brake control
- Timer functions and a real-time clock and (an optional battery is necessary)
- Fire mode
- Diagnostic and maintenance functions
 - Variety of configurable protection functions including user-defined faults
 - Supervision functions for frequency and motor variables and analog inputs
 - Motor overload protections

- An automatic reset
- Different preheat modes to prevent condensation problems
- Maintenance counters
- Drive Customizer for custom logic programming
- Control, parametrization, and diagnostics via PC-tools

1.6 Start-up Quick Guide

There are 5 applications available for the product. When an application is selected, a group of parameters get their preset values. The selection of the application makes the commissioning of the drive easy and reduces the manual work with the parameters. It is also possible to edit these parameters later.

⚠ WARNING ⚠

ACCIDENTAL START BECAUSE OF CHANGE OF SETTINGS

Running the Startup wizard, changing the application or the software can cause the I/O functions to change.

- Disconnect the motor from the drive if an accidental start can be dangerous.

The available applications:

- Standard application ([3.2 Standard and HVAC Applications](#))
- HVAC application ([3.2 Standard and HVAC Applications](#))
- PID control application ([3.3 PID Control Application](#))
- Multi-pump (single drive) application ([3.4 Multi-pump \(Single Drive\) Application](#))
- Multi-pump (double drive) application ([3.5 Multi-pump \(Multidrive\) Application](#))

Wizards make it more convenient to use the AC drive. There is a start-up wizard (see [1.6.1 Start-Up Wizard](#)), 5 applications wizards (see [4.1 Application Wizards](#)), and 1 function wizard (see [4.2.2 Fire Mode Wizard](#)).

1.6.1 Start-Up Wizard

The Start-up wizard asks for necessary data for the drive to control the procedure.

Procedure

1. Select language (P6.1).

If a battery is installed, the steps 2–5 are shown.

2. Set daylight saving time (P5.5.5). (Russia, US, EU, or OFF)
3. Set time (P5.5.2).
4. Set year (P5.5.4).
5. Set date (P5.5.3).
6. Run Startup wizard?

If the selection is No, the Start-up wizard ends.

7. Select an application (P1.2 Application, ID 212).
8. Run the Application wizard?
 - a. To continue to the application wizard, set the selection to Yes and push the [OK] button. See the description of the different application wizards in chapter [4.1 Application Wizards](#).



After these selections, the Start-up wizard is completed. To start the Start-up wizard again, there are 2 alternatives. Go to the parameter P6.5.1 Restore Factory Defaults or to the parameter B1.1.1 Start-up Wizard. Then set the value to Activate.

2 User Interfaces

2.1 Description of the Control Panel

2.1.1 Control Panel and the Keypad

The control panel is the interface between the AC drive and the user. With the control panel, it is possible to control the speed of a motor and monitor the status of the AC drive. It is also possible to set the parameters of the AC drive.

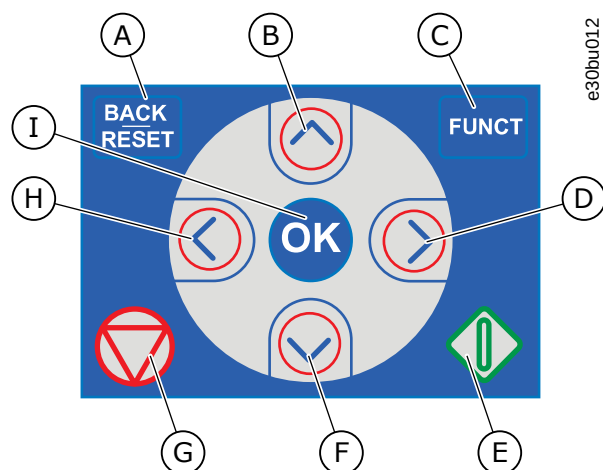


Illustration 2: Buttons of the VACON® 100 Keypad

A	The [BACK/RESET] button. Use it to move back in the menu, exit the Edit mode, reset a fault.	F	The arrow button DOWN. Use it to scroll the menu down and to decrease a value.
B	The arrow button UP. Use it to scroll up the menu and to increase a value.	G	The STOP button.
C	The [FUNCT] button. Use it to change the rotation direction of the motor, access the control page, and change the control place.	H	The arrow button LEFT. Use it to move the cursor left.
D	The arrow button RIGHT.	I	The [OK] button. Use it to go into an active level or item, or to accept a selection.
E	The START button.		

2.1.2 Displays of the Control Panel

There are 2 display types: the graphical display and the text display. The control panel always has the same keypad and buttons. The display shows this data.

- The status of the motor and the drive.
- Faults in the motor and in the drive.
- Your location in the menu structure.

If the text in the text display is too long for the display, the text scrolls to show the full text string. Some functions are only available in the graphical display.

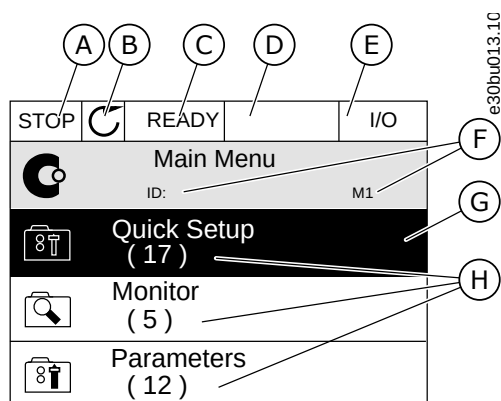


Illustration 3: Graphical Display of the Control Panel

A	The first status field: <i>STOP/RUN</i>	E	The control place field: <i>PC/I/O/KEYPAD/ FIELDBUS</i>
B	The rotation direction of the motor	F	The location field: the ID number of the parameter and the current location in the menu
C	The second status field: <i>READY/NOT READY/FAULT</i>	G	An activated group or item
D	The alarm field: <i>ALARM/-</i>	H	The number of items in the group in question

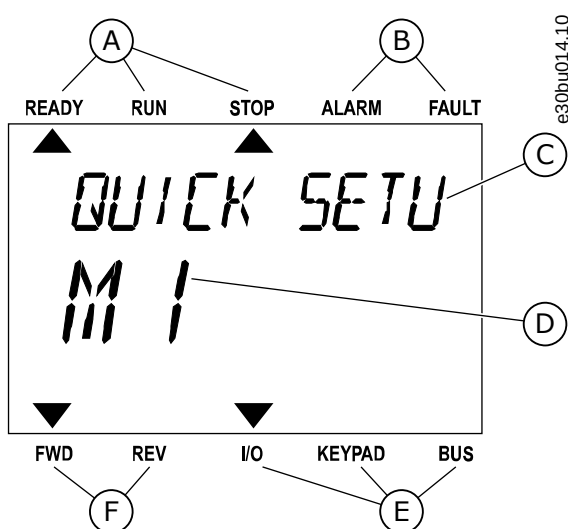


Illustration 4: Text Display of the Control Panel

A	The indicators of status	D	The current location in the menu
B	The indicators of alarm and fault	E	The indicators of the control place
C	The name of the group or item of the current location	F	The indicators of the rotation direction

2.2 Menu Structure

The data of the AC drive is in menus and submenus. To move between the menus, use the arrow buttons Up and Down in the keypad. To go into a group or an item, push the [OK] button. To go back to the previous level, push the [BACK/RESET] button.

On the display, current location in the menu shows, for example M3.2.1. The display also shows the name of the group or item in the current location.

Table 2: Menu Structure

Menu	Function
M1 Quick setup	Wizards (Content depends on P1.2 Application.)
M2 Monitor	Multimonitor
	Trend curve
	Basic
	I/O
	Temperature inputs
	Extras/Advanced
	Timer functions
	PID controller
	External PID controller
	Multi-pump
	Maintenance counters
	Fieldbus data
	Drive customizer
	Condition-based monitoring
M3 Parameters	Motor settings
	Start/Stop setup
	References
	Ramps and brakes
	I/O Configuration
	Fieldbus data
	Prohibit frequencies
	Supervisions
	Protections
	Automatic reset
	Application settings
	Timer functions
	PID controller
	External PID controller
	Multi-Pump
	Maintenance counters
	Fire mode

Menu	Function
	Motor preheat
	Drive customizer
	Pump control
	Advanced harmonic filter
	Condition-based monitoring
M4 Diagnostics	Active faults
	Reset faults
	Fault history
	Total counters
	Trip counters
	Software info
	Drive info
	Unit status
M5 I/O and Hardware	Basic I/O
	Slot C
	Slot D
	Slot E
	Real-time clock
	Power unit settings
	Keypad
	RS485
	Ethernet
	Fieldbus general
M6 User settings	Language selections
	Parameter back-up
	Drive name
	Parameter compare
	Licenses
M7 Favourites	See 9.2.1 Using the Favourites Menu .
M8 User levels	See 9.3.1 Using the User Levels Menu .

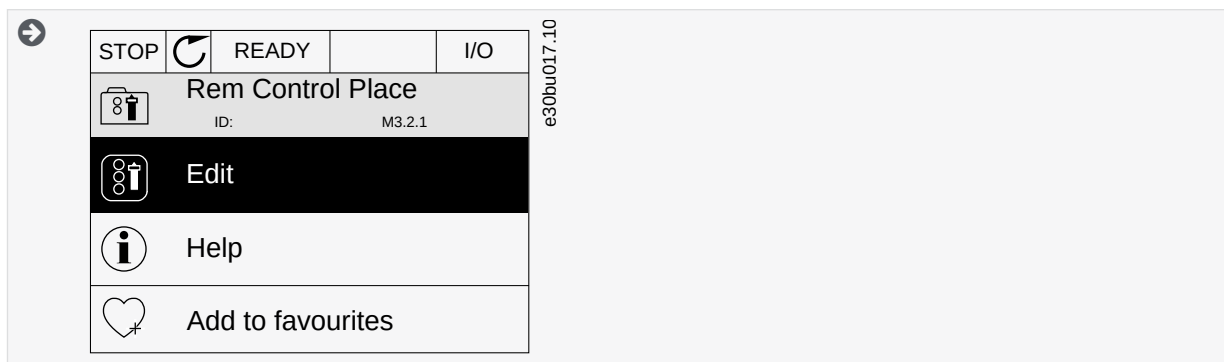
2.3 Graphical Display

2.3.1 Editing the Text Values

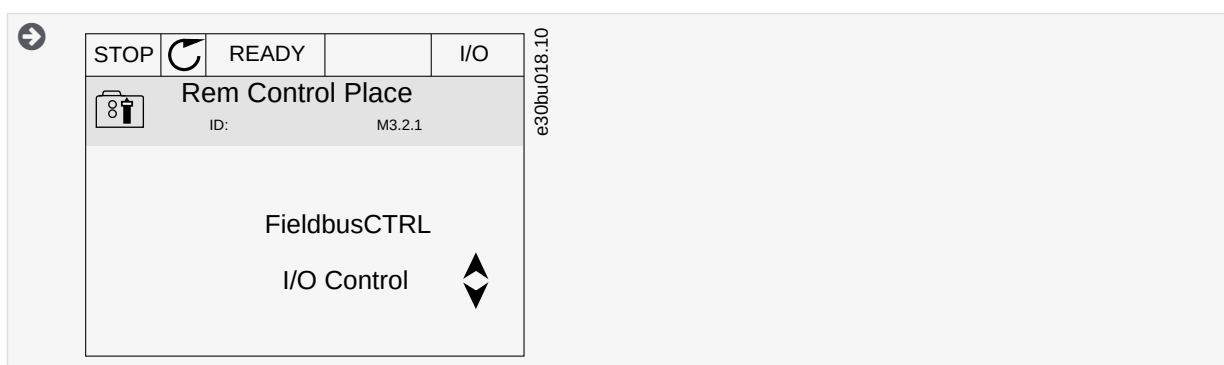
This topic gives instructions on how to edit the text values on the graphical display.

Procedure

1. Find the parameter with the arrow buttons.
2. To go to the *Edit* mode, push the [OK] button 2 times or push the arrow button Right.



3. To set a new value, push the arrow buttons Up and Down.



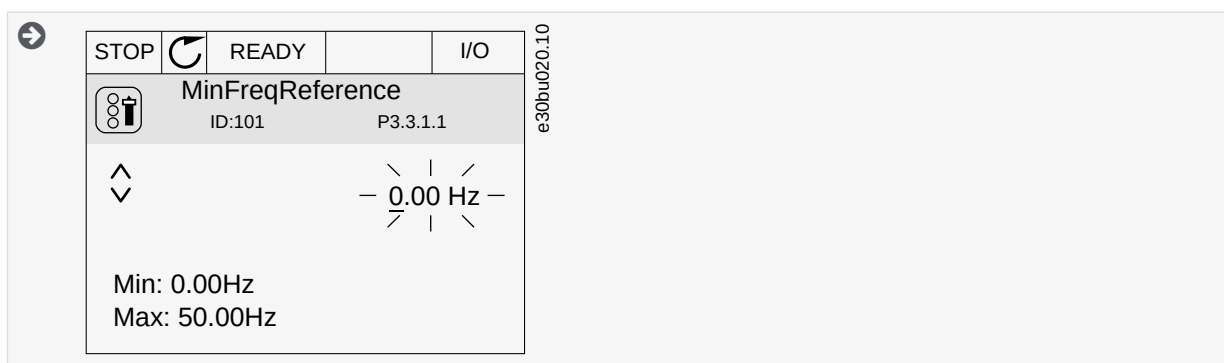
4. To accept the change, push the [OK] button. To ignore the change, use the [BACK/RESET] button.

2.3.2 Editing the Numerical Values

This topic gives instructions on how to edit the numerical values on the graphical display.

Procedure

1. Find the parameter with the arrow buttons.
2. To go to the *Edit* mode, push the [OK] button 2 times or push the arrow button Right.
3. Move from digit to digit with the arrow buttons Left and Right. Change the digits with the arrow buttons Up and Down.



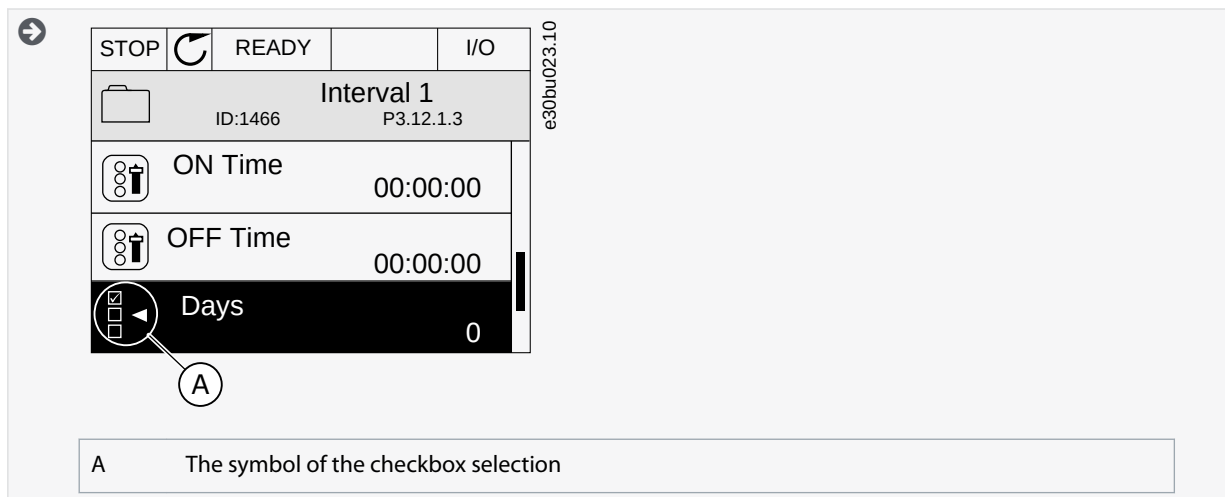
4. To accept the change, push the [OK] button. To ignore the change, use the [BACK/RESET] button.

2.3.3 Selecting More than 1 Value

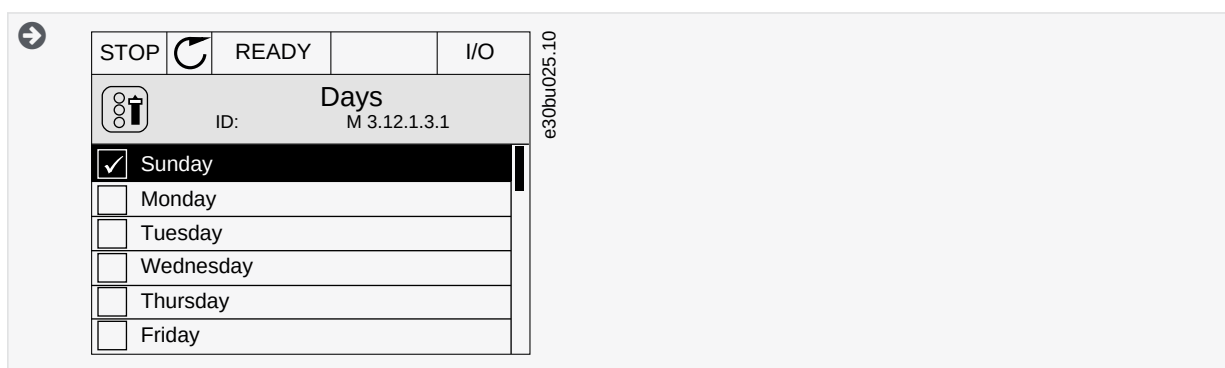
With some parameters, more than 1 value can be selected. Select a checkbox at each necessary value.

Procedure

1. Find the parameter. There is a symbol on the display when a checkbox selection is possible.



2. To move in the list of values, use the arrow buttons Up and Down.
3. To add a value into your selection, select the box that is next to it with the arrow button Right.



2.3.4 Resetting a Fault

To reset a fault, use the [BACK/RESET] button or the parameter Reset Faults. See [7.1.2 Reset Faults](#).

2.3.5 [FUNCT] Button

The [FUNCT] button can be used for 4 functions.

- To have access to the Control page.
- To change easily between the Local and Remote control places.
- To change the rotation direction.
- To edit quickly a parameter value.

2.3.5.1 Different Control Places

The selection of the control place determines from where the AC drive takes the start and stop commands. All the control places have a parameter for the selection of the frequency reference source. The Local control place is always *Keypad*. The Remote control place is *I/O* or *Fieldbus*. The status bar of the display shows the current control place.

It is possible to use *I/O A*, *I/O B*, and *Fieldbus* as Remote control places. *I/O A* and *Fieldbus* have the lowest priority. Select them with P3.2.1 (Remote Control Place). *I/O B* can bypass the Remote control places *I/O A* and *Fieldbus* with a digital input. Select the digital input with parameter P3.5.1.7 (I/O B Control Force).

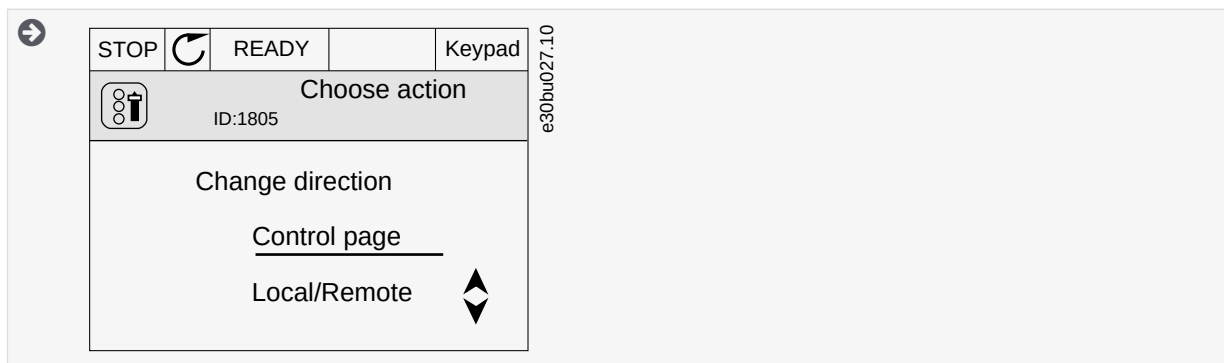
Keypad is always used as a control place when the control place is Local. Local control has higher priority than Remote control. For example, if parameter P3.5.1.7 bypasses the control place with a digital input, and Local is selected in Remote control, *Keypad* becomes the control place. Use the [FUNCT] button or P3.2.2 Local/Remote to change between the Local and Remote control.

2.3.5.2 Changing the Control Place

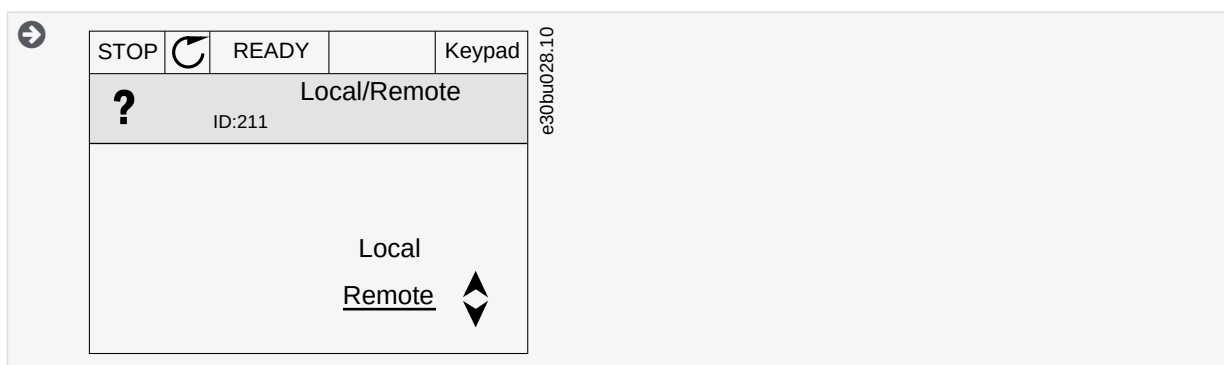
This topic gives instructions on how to change the control place on the control panel.

Procedure

1. Anywhere in the menu structure, push the [FUNCT] button.
2. To select Local/Remote, use the arrow buttons Up and Down. Push the [OK] button.



3. To select Local or Remote, use the arrow buttons Up and Down again. To accept the selection, push the [OK] button.



4. If Remote control place is changed to Local, that is, *Keypad*, give a keypad reference.

➡ After the selection, the display goes back into the same location where it was when the [FUNCT] button was pushed.

2.3.5.3 Going into the Control Page

It is easy to monitor the most important values in the *Control page*.

See more information about Keypad Reference in [6.3 Group 3.3: References](#). If other control places or reference values are used, the display shows the frequency reference, which cannot be edited. The other values on the page are *Multimonitor* values. These values can be selected (see [5.2.1 Changing the Items to Monitor](#)).

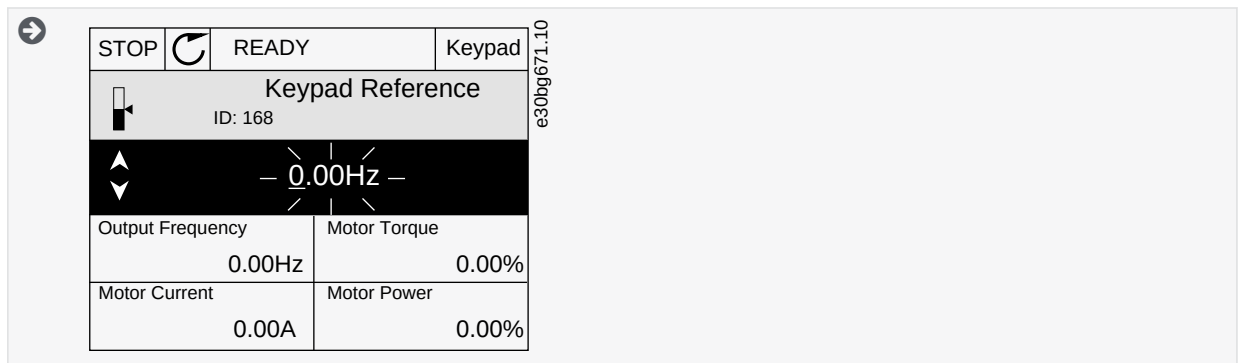
Procedure

1. Anywhere in the menu structure, push the [FUNCT] button.
2. To select the *Control page*, push the arrow buttons Up and Down. Go in with the [OK] button.

➡ The *Control page* opens.

3. With the Local control place and the keypad reference selected, P3.3.1.8 Keypad Reference can be set with the [OK] button.

- To change the digits in the value, push the arrow buttons Up and Down. Accept the change with the [OK] button.



2.3.5.4 Changing the Rotation Direction

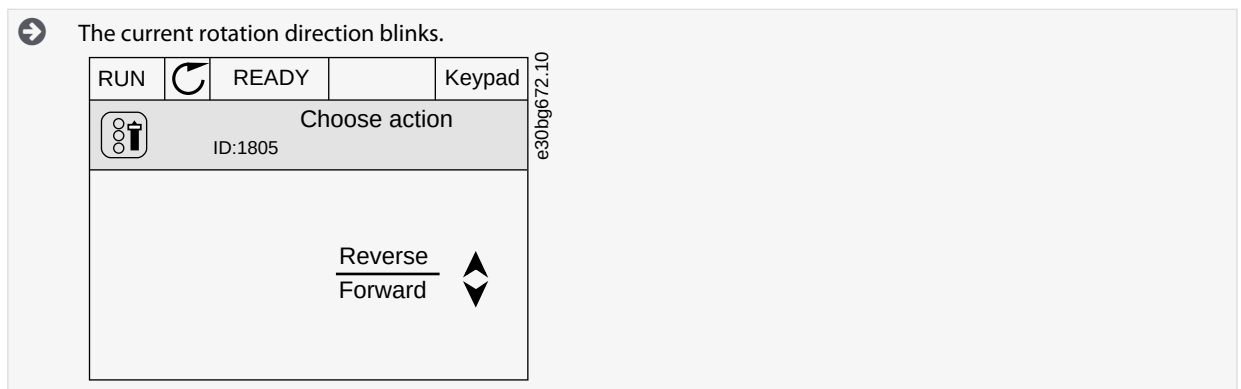
The rotation direction of the motor can be changed quickly with the [FUNCT] button.

NOTICE

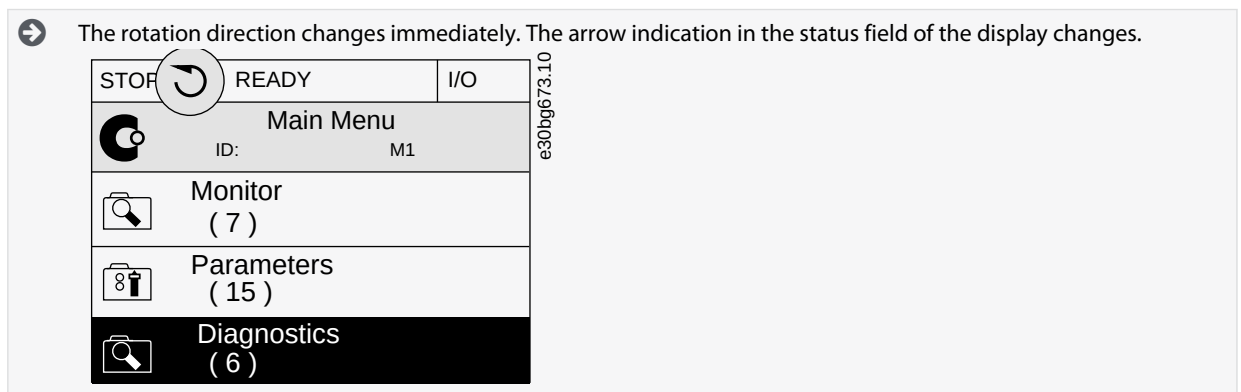
The command *Change direction* is available in the menu only if the current control place is Local.

Procedure

- Anywhere in the menu structure, push the [FUNCT] button.
- To select *Change direction*, push the arrow buttons Up and Down. Push the [OK] button.
- Select a new rotation direction.



- Push the [OK] button.



2.3.5.5 Using the Quick Edit Function

With the *Quick Edit* function, a parameter can be accessed quickly by typing the ID number of the parameter.

Procedure

- Anywhere in the menu structure, push the [FUNCT] button.

2. Push the arrow buttons Up and Down to select *Quick Edit* and accept with the [OK] button.
3. Write the ID number of a parameter or monitoring value. Push [OK].

➡ The display shows the parameter value in the edit mode and the monitoring value in the monitoring mode.

2.3.6 Copying the Parameters of an AC Drive

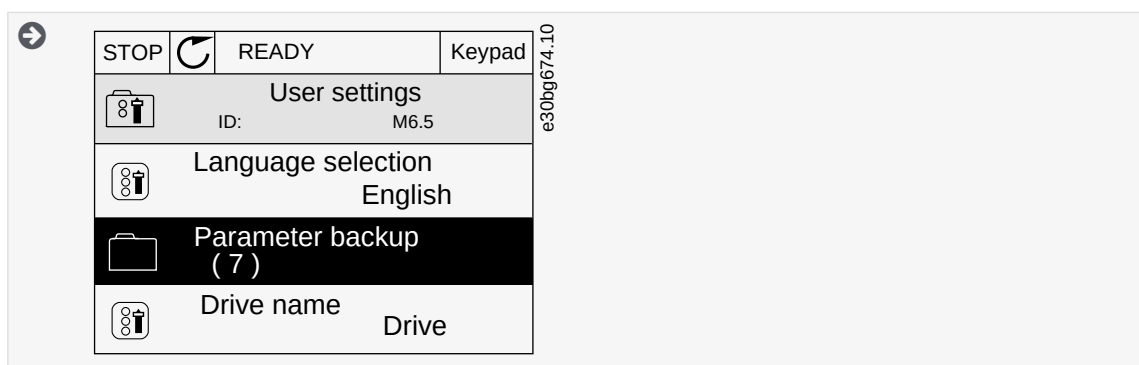
Use this function to copy parameters from a drive to another. Before downloading parameters from the control panel to the drive, stop the drive.

NOTICE

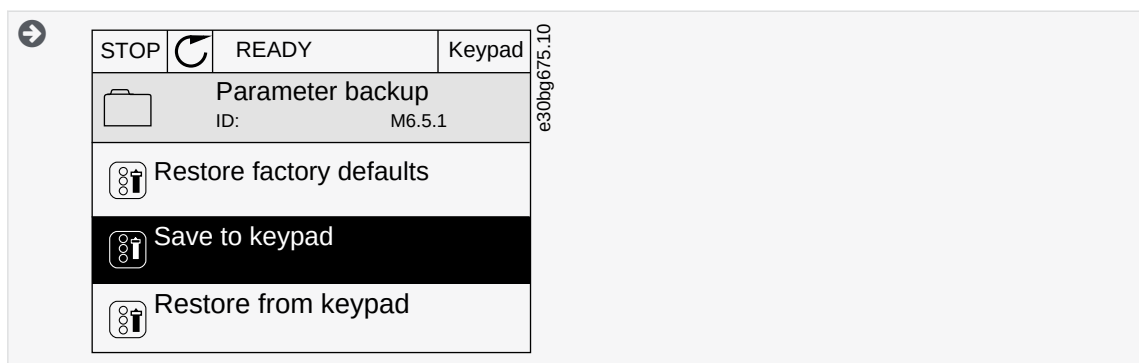
This function is available only in the graphical display.

Procedure

1. Save the parameters to the control panel.
 - a. Go into the *User settings* menu.
 - b. Go into the *Parameter backup* submenu.



- c. Use the arrow buttons Up and Down to select a function. Accept the selection with the [OK] button.



The command *Restore factory defaults* brings back the parameter settings that were made at the factory. Use the command *Save to keypad* to copy all the parameters to the control panel. The command *Restore from keypad* copies all the parameters from the control panel to the drive.

2. Detach the control panel and connect it to another drive.
3. Download the parameters to the new drive with the command *Restore from keypad*.

2.3.7 Comparing the Parameters

Use this function to compare the current parameter set with one of these 4 sets. See more about these parameters in [9.1.2 Parameter Back-up Parameters](#).

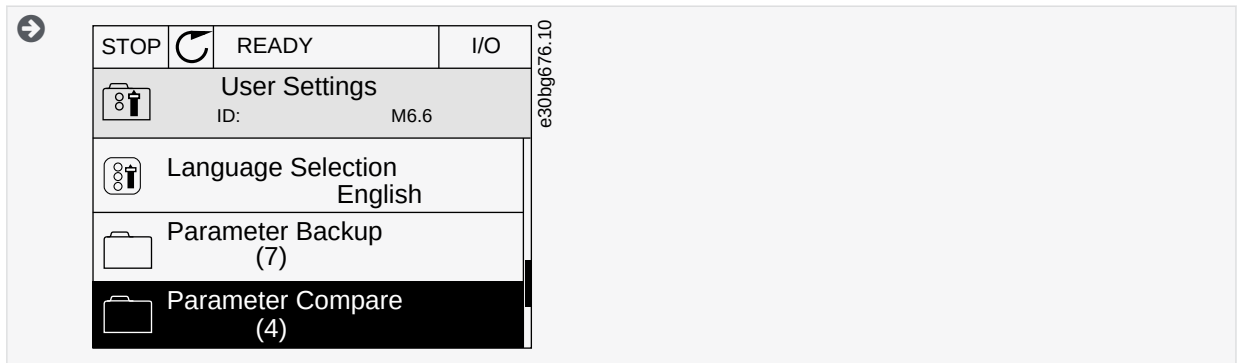
- *Set 1* (P6.5.4 Save to Set 1)
- *Set 2* (P6.5.6 Save to Set 2)
- *Defaults* (P6.5.1 Restore Factory Defaults)
- *Keypad set* (P6.5.2 Save to Keypad)

N O T I C E

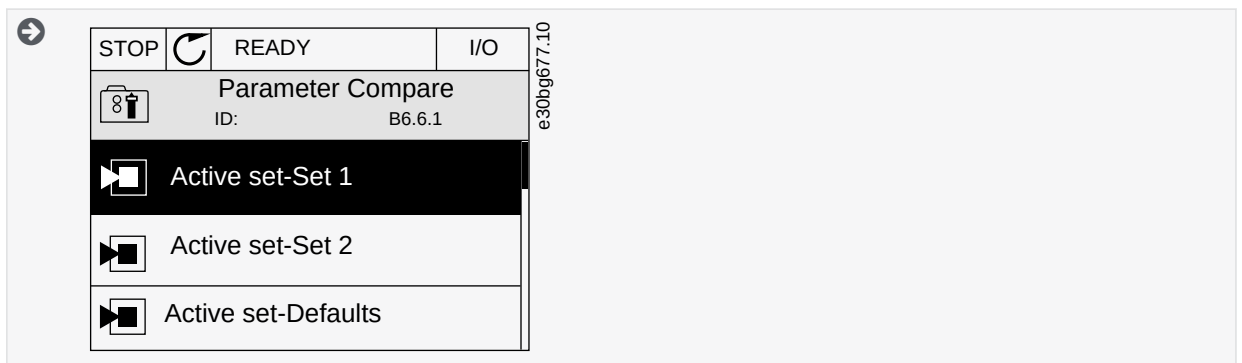
If the parameter set with which the current set is compared was not saved, the display shows the text *Comparing failed*.

Procedure

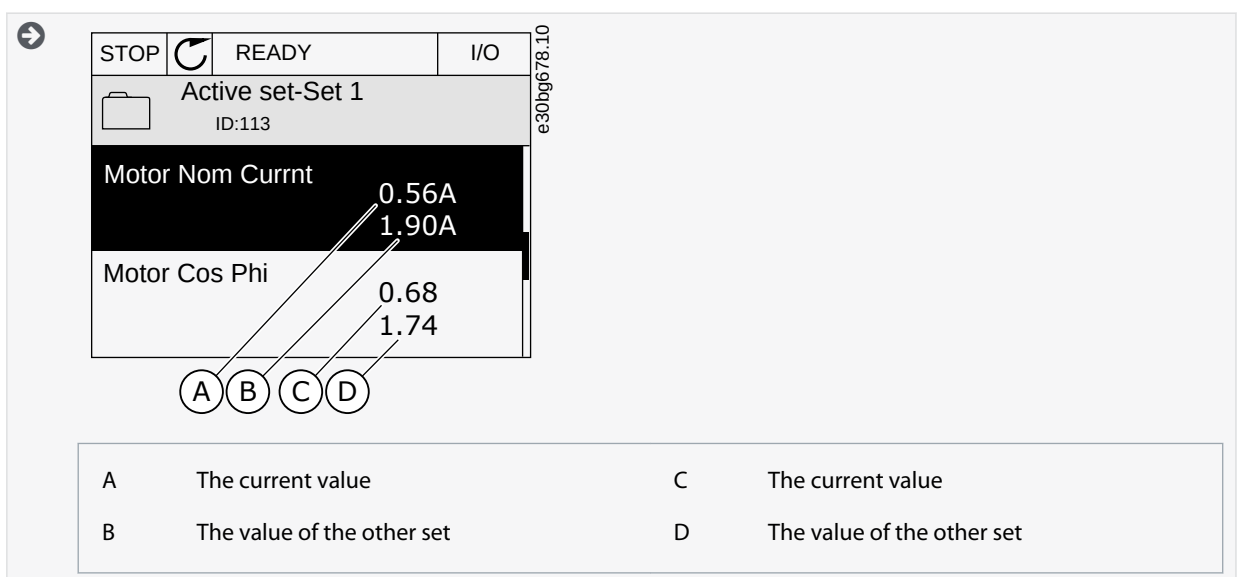
1. Go into the *User settings* menu.
2. Go into the *Parameter Compare* function.



3. Select the pair of sets. Push [OK] to accept the selection.



4. Select *Active* and push [OK].
5. Examine the comparing between the current values and the values of the other set.



2.3.8 Reading Help Texts

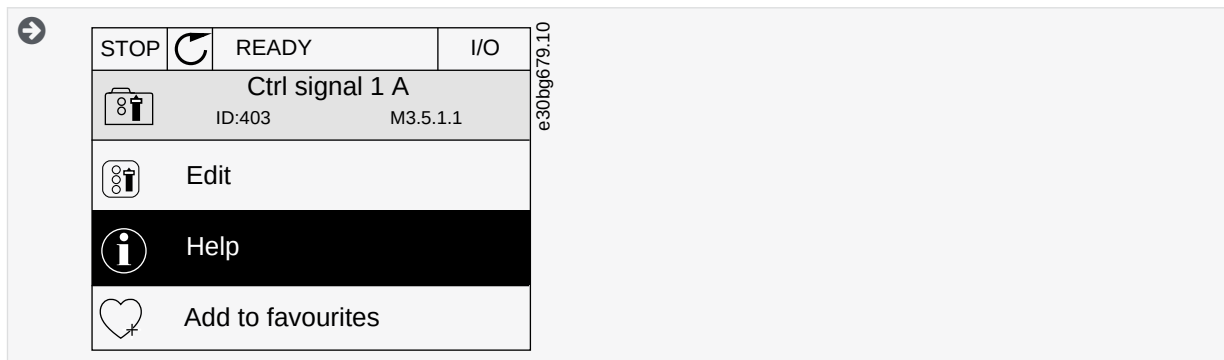
The graphical display can show help texts on many topics. All the parameters have a help text. The help texts are also available for the faults, alarms, and the start-up wizard.

N O T I C E

The help texts are always in English.

Procedure

1. Find the item to read.
2. Use the arrow buttons Up and Down to select *Help*.



3. To open the help text, push the [OK] button.

2.3.9 Using the Favourites Menu

If the same items are used frequently, add them into *Favourites*. Collect a set of parameters or monitoring signals from all the control panel menus. See [9.2.2 Adding an Item to the Favourites](#).

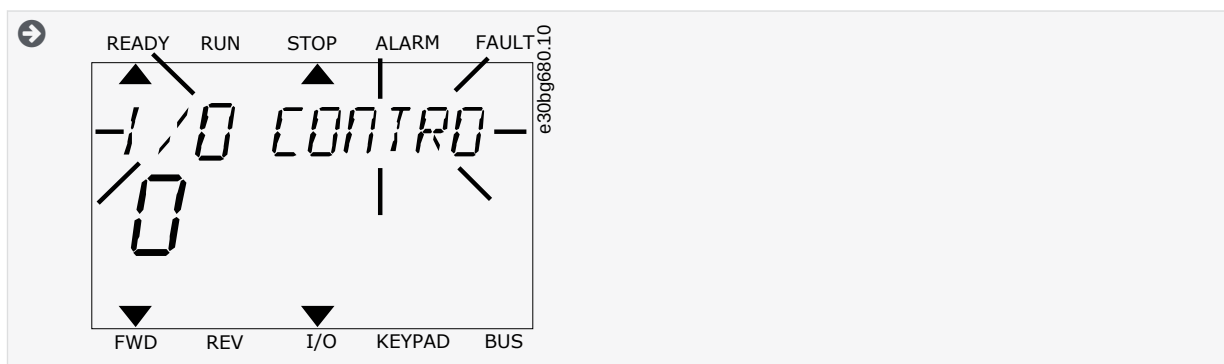
2.4 Text Display

2.4.1 Editing the Text Values

This topic gives instructions on how to edit text values on the text display.

Procedure

1. Find the parameter with the arrow buttons.
2. To go to the Edit mode, push the [OK] button.
3. To set a new value, push the arrow buttons Up and Down.



4. Accept the change with the [OK] button. To ignore the change, use the [BACK/RESET] button.

2.4.2 Editing the Numerical Values

This topic gives instructions on how to edit the numerical values on the text display.

Procedure

1. Find the parameter with the arrow buttons.
2. Go to the Edit mode.
3. Move from digit to digit with the arrow buttons Left and Right. Change the digits with the arrow buttons Up and Down.
4. Accept the change with the [OK] button. To ignore the change, use the [BACK/RESET] button.

2.4.3 Resetting a Fault

To reset a fault, use the [BACK/RESET] button or the parameter Reset Faults. See [7.1.2 Reset Faults](#).

2.4.4 [FUNCT] Button

The [FUNCT] button can be used for 4 functions.

- To have access to the *Control page*.
- To change easily between the Local and Remote control places.
- To change the rotation direction.
- To edit quickly a parameter value.

2.4.4.1 Different Control Places

The selection of the control place determines from where the AC drive takes the start and stop commands. All the control places have a parameter for the selection of the frequency reference source. The Local control place is always *Keypad*. The Remote control place is *I/O* or *Fieldbus*. The status bar of the display shows the current control place.

It is possible to use *I/O A*, *I/O B*, and *Fieldbus* as Remote control places. *I/O A* and *Fieldbus* have the lowest priority. Select them with P3.2.1 (Remote Control Place). *I/O B* can bypass the Remote control places *I/O A* and *Fieldbus* with a digital input. Select the digital input with parameter P3.5.1.7 (*I/O B* Control Force).

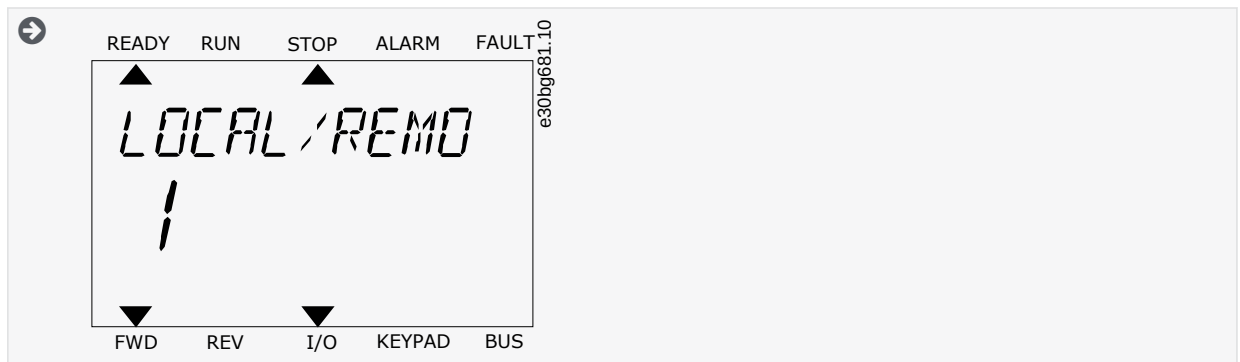
Keypad is always used as a control place when the control place is Local. Local control has higher priority than Remote control. For example, if parameter P3.5.1.7 bypasses the control place with a digital input, and Local is selected in Remote control, *Keypad* becomes the control place. Use the [FUNCT] button or P3.2.2 Local/Remote to change between the Local and Remote control.

2.4.4.2 Changing the Control Place

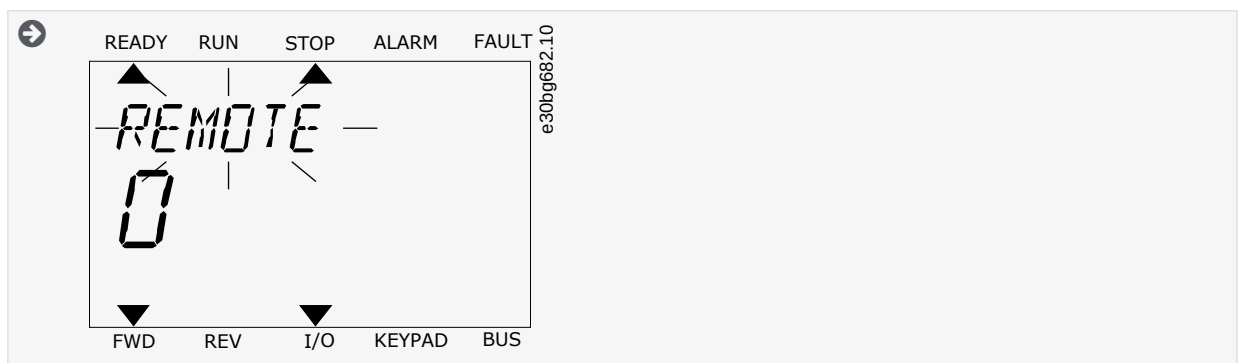
This topic gives instructions on how to change the control place on the control panel.

Procedure

1. Anywhere in the menu structure, push the [FUNCT] button.
2. To select *Local/Remote*, use the arrow buttons Up and Down. Push the [OK] button.



3. To select *Local* or *Remote*, use the arrow buttons Up and Down again. To accept the selection, push the [OK] button.



4. If *Remote* control place is changed to *Local*, that is, the keypad, give a keypad reference.

➡ After the selection, the display goes back into the same location where it was when the [FUNCT] button was pushed.

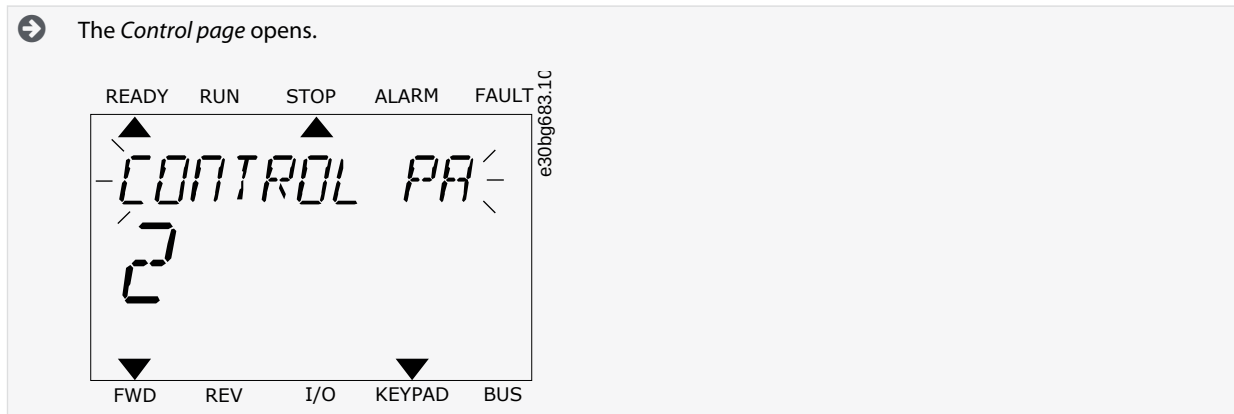
2.4.4.3 Going into the Control Page

It is easy to monitor the most important values in the *Control page*.

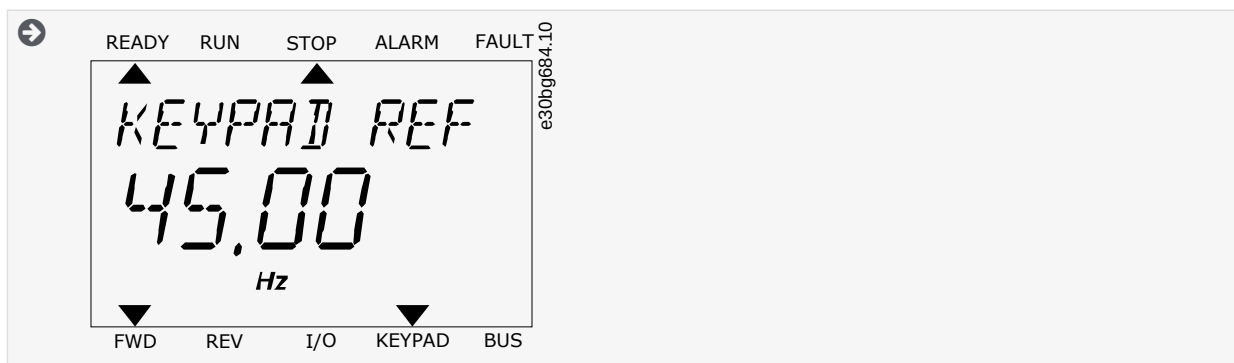
See more information about Keypad Reference in [6.3 Group 3.3: References](#). If other control places or reference values are used, the display shows the frequency reference that cannot be edited. The other values on the page are *Multimonitor* values. These values can be selected (see [5.2.1 Changing the Items to Monitor](#)).

Procedure

1. Anywhere in the menu structure, push the [FUNCT] button.
2. To select the *Control page*, push the arrow buttons Up and Down. Go in with the [OK] button.



3. If the Local control place and the keypad reference are used, P3.3.1.8 Keypad Reference can be set with the [OK] button.



2.4.4.4 Changing the Rotation Direction

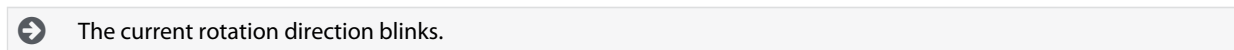
The rotation direction of the motor can be changed quickly with the [FUNCT] button.

NOTICE

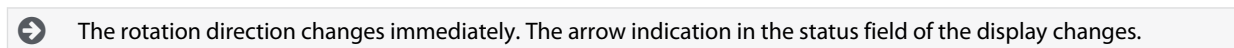
The command *Change direction* is available in the menu only if the current control place is Local.

Procedure

1. Anywhere in the menu structure, push the [FUNCT] button.
2. To select *Change direction*, push the arrow buttons Up and Down. Push the [OK] button.
3. Select a new rotation direction.



4. Push the [OK] button.



2.4.4.5 Using the Quick Edit Function

With the *Quick Edit* function, a parameter can be accessed quickly by typing the ID number of the parameter.

Procedure

1. Anywhere in the menu structure, push the [FUNCT] button.
2. Push the arrow buttons Up and Down to select *Quick Edit* and accept with the [OK] button.
3. Write the ID number of a parameter or monitoring value. Push [OK].

The display shows the parameter value in the edit mode and the monitoring value in the monitoring mode.

2.5 VACON® Live PC Tool

VACON® Live is a PC tool for commissioning and maintenance of the VACON® AC drives. VACON® Live can be downloaded from <https://www.danfoss.com>.

The VACON® Live PC tool includes these functions.

- Parameterization, monitoring, drive info, data logger, and so on.
- The software download tool VACON® Loader
- Serial communication and Ethernet support
- Windows XP, Vista, 7, 8, and 10 support
- 17 languages: English, German, Spanish, Finnish, French, Italian, Russian, Swedish, Chinese, Czech, Danish, Dutch, Polish, Portuguese, Romanian, Slovak and Turkish

Make the connection between the AC drive and the PC tool with the VACON® serial communication cable. The serial communication drivers are installed automatically during the installation of VACON® Live. After the cable is installed, VACON® Live finds the connected drive automatically.

See more on how to use VACON® Live in the help menu of the program.

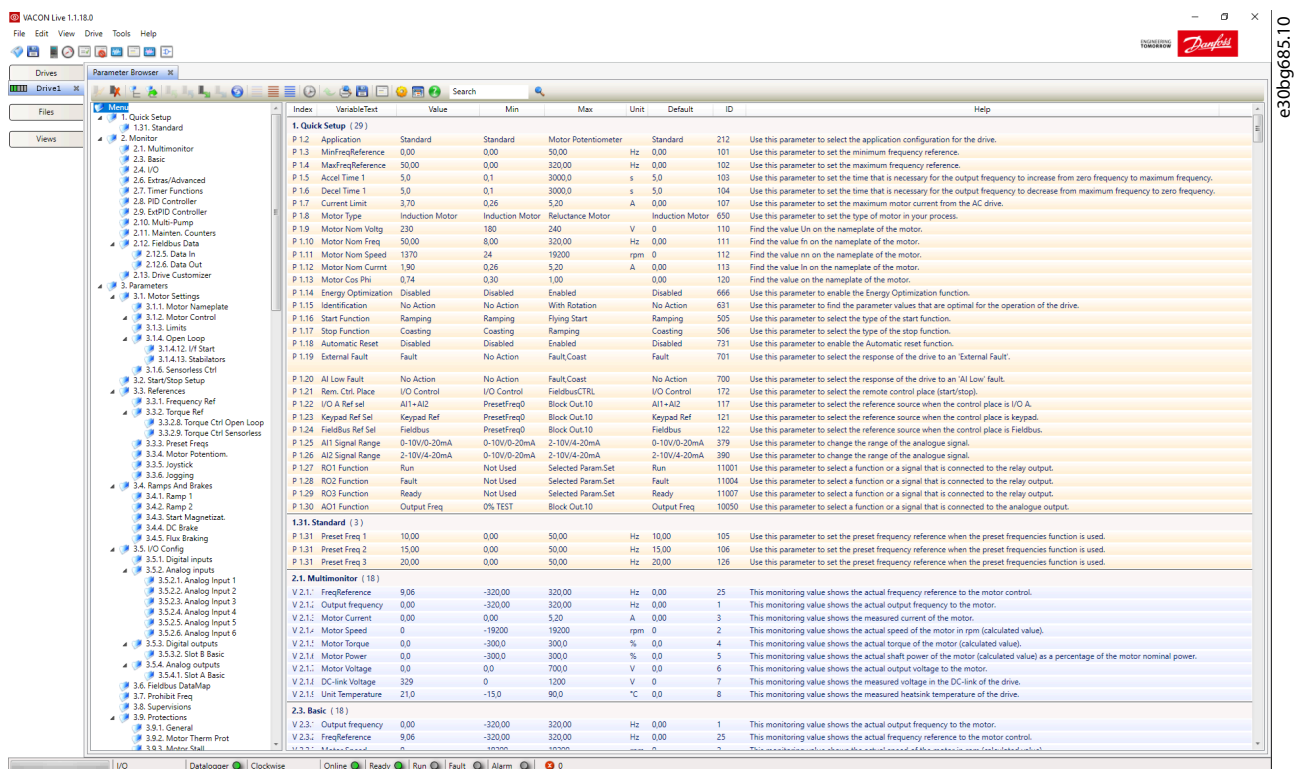


Illustration 5: The VACON® Live PC Tool

3 Applications

3.1 Using the Applications

Use the parameter P1.2 (Application) to select an application for the drive. Immediately when the parameter P1.2 changes, parameter groups get their preset values.

3.2 Standard and HVAC Applications

Use the Standard and HVAC applications to control pumps or fans, for example.

It is possible to control the drive from the control panel (keypad), Fieldbus, or I/O terminal.

When the drive is controlled with the I/O terminal, the frequency reference signal is connected to AI1 (0...10 V) or AI2 (4...20 mA).

The connection depends the type of the signal. There are also 3 preset frequency references available. The preset frequency references can be activated with DI4 and DI5. The start/stop signals of the drive are connected to DI1 (start forward) and DI2 (start reverse).

It is possible to configure all the drive outputs freely in all the applications. There is 1 analog output (Output Frequency) and 3 relay outputs (Run, Fault, Ready) available on the basic I/O board.

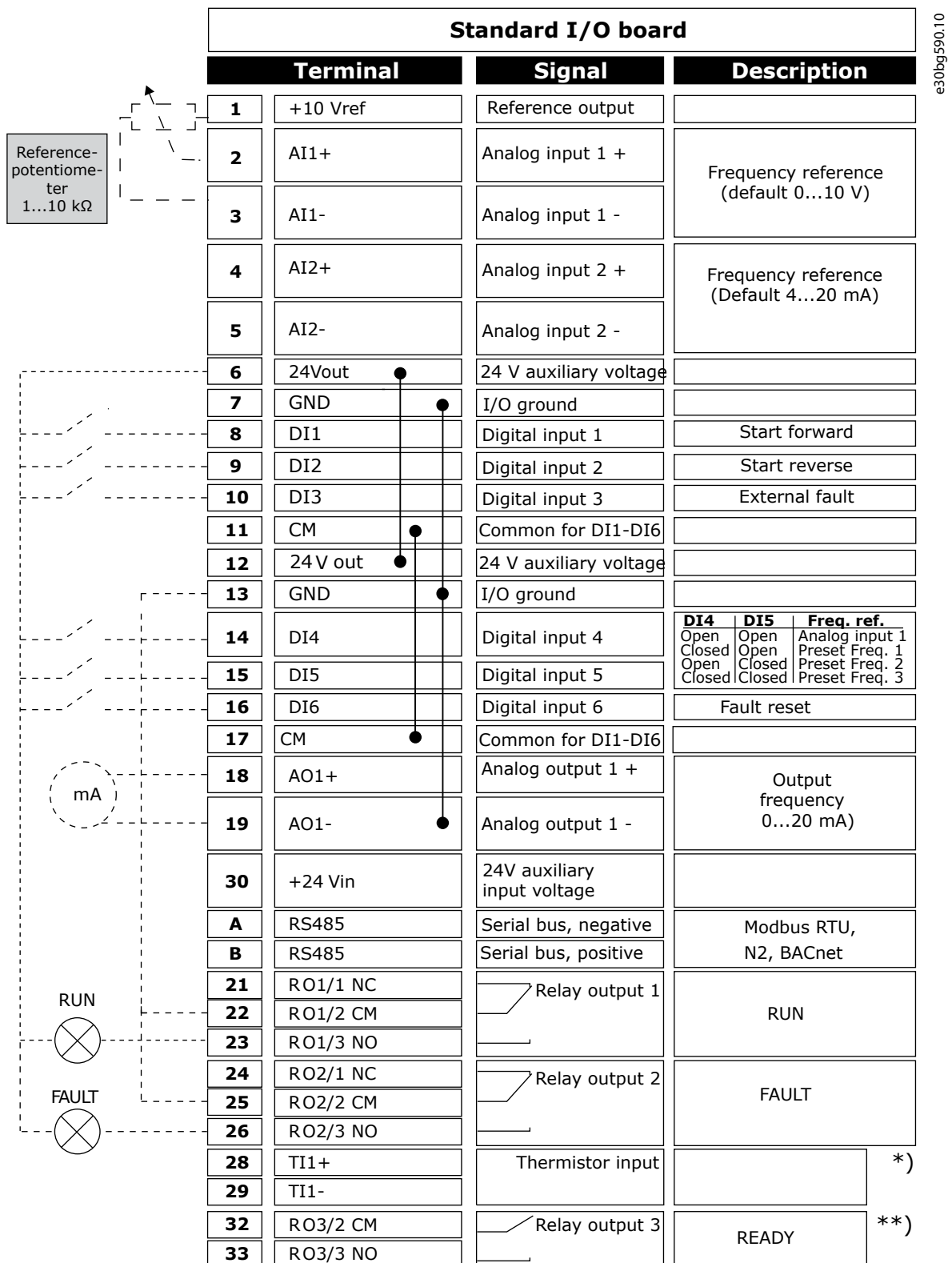


Illustration 6: The Default Control Connections of the Standard and HVAC Applications

*) Available only in VACON® 100 X. **) For the DIP switch configurations in VACON® 100 X, see the VACON® 100 X Installation manual.

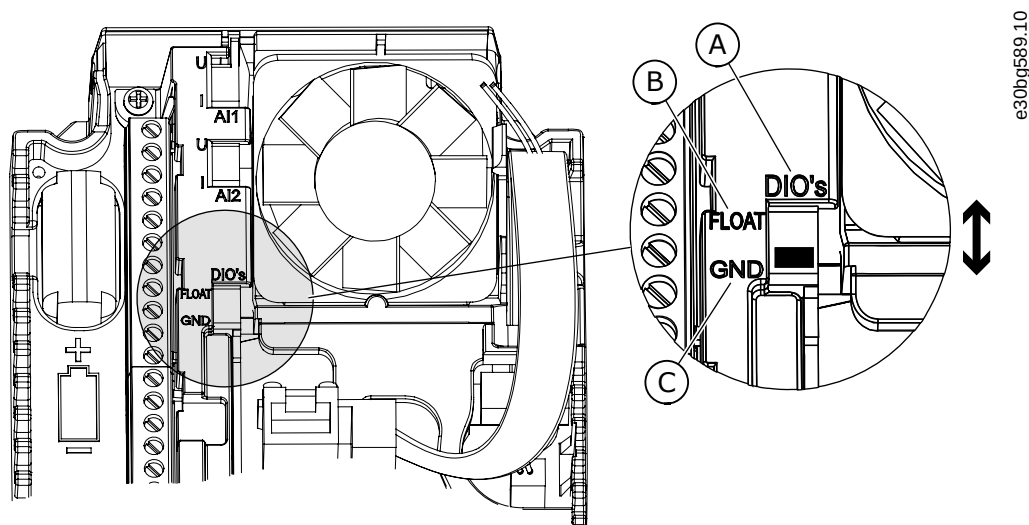


Illustration 7: The DIP Switch

A	Digital input DIP switch	C	Connected to GND (Default)
B	Floating		

See information on the Startup, and Fire mode wizards in [4.2.1 Wizards in Menu 1.1](#).

Table 3: M1 Quick Setup

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.2	Application	0	4		0	212	0 = Standard 1 = HVAC 2 = PID Control 3 = Multi-pump (single drive) 4 = Multi-pump (multidrive)
1.3	Minimum Frequency Reference	0.00	P1.4	Hz	0.0	101	
1.4	Maximum Frequency Reference	P1.3	320.0	Hz	50.0/60.0	102	
1.5	Acceleration Time 1	0.1	3000.0	s	5.0	103	
1.6	Deceleration Time 1	0.1	3000.0	s	5.0	104	
1.7	Motor Current Limit	I _H *0.1	I _S	A	Varies	107	
1.8	Motor Type	0	2		0	650	0 = Induction Motor 1 = Permanent Magnet Motor 2 = Reluctance Motor
1.9	Motor Nominal Voltage	Varies	Varies	V	Varies	110	Find this value U _n on the nameplate of the motor. NOTE! Find out if the motor connection is Delta or Star.
1.10	Motor Nominal Frequency	8.0	320.0	Hz	50/60	111	Find this value f _n on the nameplate of the motor.

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.11	Motor Nominal Speed	24	19200	RPM	Varies	112	Find this value n_n on the nameplate of the motor.
1.12	Motor Nominal Current	IH * 0.1	IH * 2	A	Varies	113	Find this value I_n on the nameplate of the motor.
1.13	Motor Cos Phi (Power Factor)	0.30	1.00		Varies	120	Find this value on the nameplate of the motor.
1.14	Energy Optimization	0	1		0	666	0 = Disabled 1 = Enabled
1.15	Identification	0	2		0	631	0 = No action 1 = At standstill 2 = With rotation
1.16	Start Function	0	1		0	505	0 = Ramping 1 = Flying Start
1.17	Stop Function	0	1		0	506	0 = Coasting 1 = Ramping
1.18	Automatic Reset	0	1		0	731	0 = Disabled 1 = Enabled
1.19	Response to External Fault	0	3		2	701	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)
1.20	Response to AI Low Fault	0	5		0	700	0 = No action 1 = Alarm 2 = Alarm+preset fault frequency (P3.9.1.13) 3 = Alarm + previous frequency 4 = Fault (Stop according to stop mode) 5 = Fault (Stop by coasting)
1.21	Remote Control Place	0	1		0	172	0 = I/O control 1 = Fieldbus control
1.22	I/O Control Reference A Selection	1	20		5	117	1 = Preset Frequency 0 2 = Keypad reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID reference 8 = Motor potentiometer 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5

Index	Parameter	Min	Max	Unit	Default	ID	Description
							16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10
1.23	Keypad Control Reference Selection	1	20		1	121	See P1.22.
1.24	Fieldbus Control Reference Selection	1	20		2	122	See P1.22.
1.25	AI1 Signal Range	0	1		0	379	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.26	AI2 Signal Range	0	1		1	390	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.27	RO1 Function	0	74		2	11001	See P3.5.3.2.1.
1.28	RO2 Function	0	74		3	11004	See P3.5.3.2.1.
1.29	RO3 Function	0	74		1	11007	See P3.5.3.2.1.
1.30	AO1 Function	0	31		2	10050	See P3.5.4.1.1.

Table 4: M1.31 Standard/M1.32 HVAC

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.31.1	Preset Frequency 1	P1.3	P1.4	Hz	10.0	105	
1.31.2	Preset Frequency 2	P1.3	P1.4	Hz	15.0	106	
1.31.3	Preset Frequency 3	P1.3	P1.4	Hz	20.0	126	

3.3 PID Control Application

Use the PID control application with processes where the process variable (for example pressure) is controlled through control of the speed of the motor.

In this application, the internal PID controller of the drive is configured for 1 setpoint and 1 feedback signal.

It is possible to use 2 control places. Make the selection of the control place A or B with DI6. When control place A is active, the start/stop commands are given by DI1, and the PID controller gives the frequency reference. When control place B is active, start/stop commands are given by DI4, and AI1 gives the frequency reference.

It is possible to configure all the drive outputs freely in all the applications. There are 1 analog output (Output Frequency) and 3 relay outputs (Run, Fault, Ready) available on the basic I/O board.

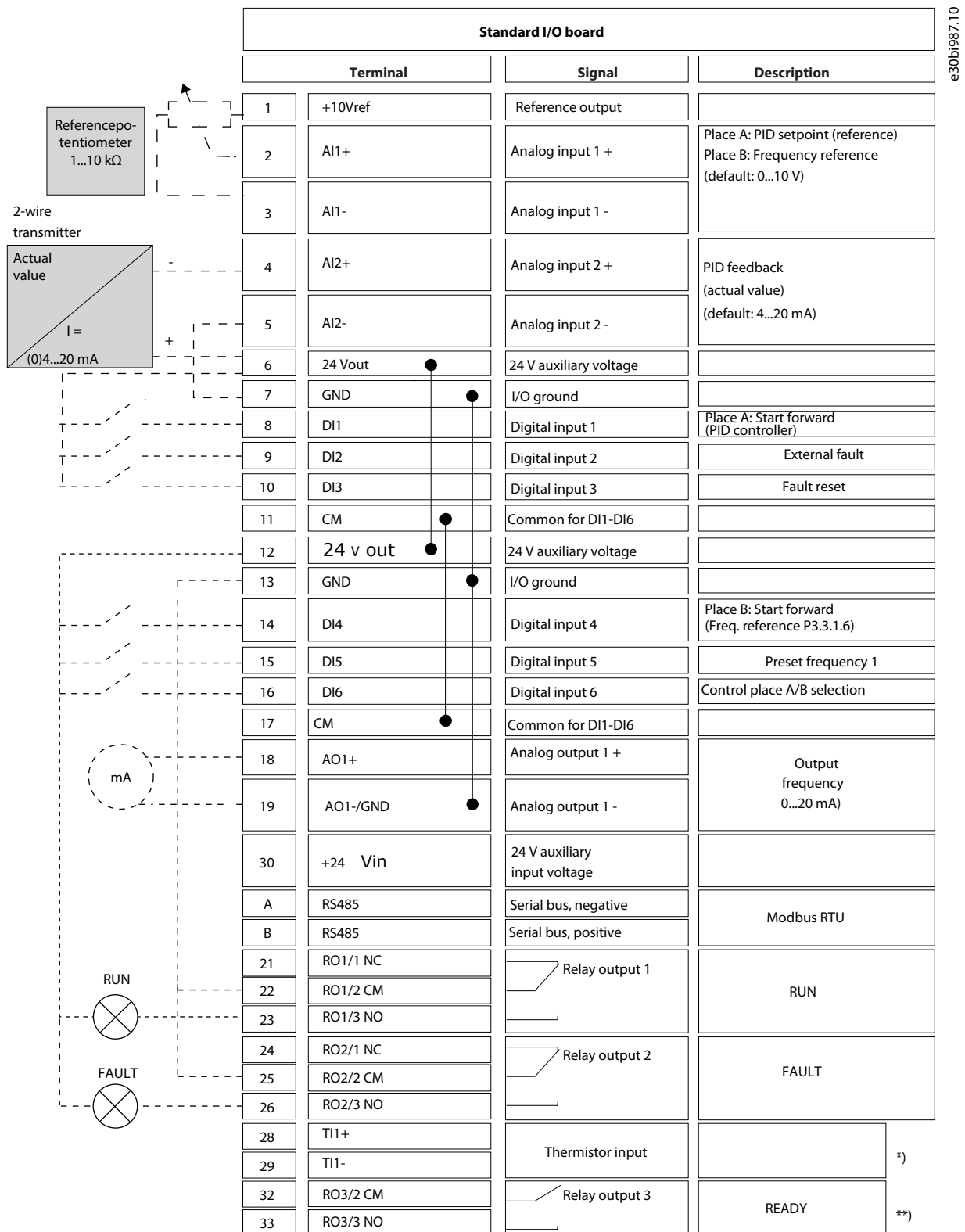


Illustration 8: The Default Control Connections of the PID Control Application

*) Available only in VACON® 100 X. **) For the DIP switch configurations in VACON® 100 X, see the VACON® 100 X Installation manual.

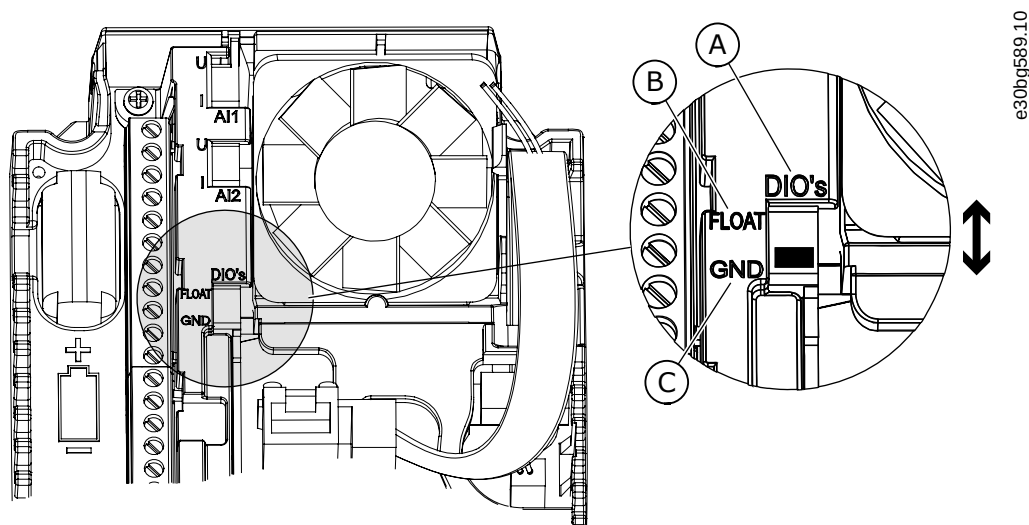


Illustration 9: The DIP Switch

A	Digital input DIP switch	C	Connected to GND (Default)
B	Floating		

See information on the Startup, and Fire mode wizards in [4.2.1 Wizards in Menu 1.1](#).

Table 5: M1 Quick Setup

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.2	Application	0	4		2	212	0 = Standard 1 = HVAC 2 = PID Control 3 = Multi-pump (single drive) 4 = Multi-pump (multidrive)
1.3	Minimum Frequency Reference	0.00	P1.4	Hz	0.0	101	
1.4	Maximum Frequency Reference	P1.3	320.0	Hz	50.0/60.0	102	
1.5	Acceleration Time 1	0.1	3000.0	s	5.0	103	
1.6	Deceleration Time 1	0.1	3000.0	s	5.0	104	
1.7	Motor Current Limit	I _H *0.1	I _S	A	Varies	107	
1.8	Motor Type	0	2		0	650	0 = Induction Motor 1 = Permanent Magnet Motor 2 = Reluctance Motor
1.9	Motor Nominal Voltage	Varies	Varies	V	Varies	110	Find this value U _n on the nameplate of the motor. NOTE! Find out if the motor connection is Delta or Star.
1.10	Motor Nominal Frequency	8.0	320.0	Hz	50/60	111	Find this value f _n on the nameplate of the motor.

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.11	Motor Nominal Speed	24	19200	RPM	Varies	112	Find this value n_n on the nameplate of the motor.
1.12	Motor Nominal Current	IH * 0.1	IH * 2	A	Varies	113	Find this value I_n on the nameplate of the motor.
1.13	Motor Cos Phi (Power Factor)	0.30	1.00		Varies	120	Find this value on the nameplate of the motor.
1.14	Energy Optimization	0	1		0	666	0 = Disabled 1 = Enabled
1.15	Identification	0	2		0	631	0 = No action 1 = At standstill 2 = With rotation
1.16	Start Function	0	1		0	505	0 = Ramping 1 = Flying Start
1.17	Stop Function	0	1		0	506	0 = Coasting 1 = Ramping
1.18	Automatic Reset	0	1		0	731	0 = Disabled 1 = Enabled
1.19	Response to External Fault	0	3		2	701	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)
1.20	Response to AI Low Fault	0	5		0	700	0 = No action 1 = Alarm 2 = Alarm+preset fault frequency (P3.9.1.13) 3 = Alarm + previous frequency 4 = Fault (Stop according to stop mode) 5 = Fault (Stop by coasting)
1.21	Remote Control Place	0	1		0	172	0 = I/O control 1 = Fieldbus control
1.22	I/O Control Reference A Selection	1	20		6	117	1 = Preset Frequency 0 2 = Keypad reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID reference 8 = Motor potentiometer 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5

Index	Parameter	Min	Max	Unit	Default	ID	Description
							16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10
1.23	Keypad Control Reference Selection	1	20		1	121	See P1.22.
1.24	Fieldbus Control Reference Selection	1	20		2	122	See P1.22.
1.25	AI1 Signal Range	0	1		0	379	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.26	AI2 Signal Range	0	1		1	390	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.27	RO1 Function	0	74		2	11001	See P3.5.3.2.1.
1.28	RO2 Function	0	74		3	11004	See P3.5.3.2.1.
1.29	RO3 Function	0	74		1	11007	See P3.5.3.2.1.
1.30	AO1 Function	0	31		2	10050	See P3.5.4.1.1.

Table 6: M1.33 PID Control

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.33.1	PID Gain	0.00	100.00	%	100.00	118	
1.33.2	PID Integration Time	0.00	600.00	s	1.00	119	
1.33.3	PID Derivation Time	0.00	100.00	s	0.00	132	
1.33.4	Process Unit Selection	1	44		1	1036	See P3.13.1.4
1.33.5	Process Unit Min	Varies	Varies		Varies	1033	See P3.13.2.6.
1.33.6	Process Unit Max	Varies	Varies		Varies	1034	
1.33.7	Feedback 1 Source Selection	0	30		2	334	See P3.13.3.3
1.33.8	Setpoint 1 Source Selection	0	32		1	332	See P3.13.2.6
1.33.9	Keypad Setpoint 1	Varies	Varies	Varies	0	167	
1.33.10	SP1 Sleep Frequency Limit	0.0	320.0	Hz	0.0	1016	
1.33.11	SP1 Sleep Delay	0	3000	s	0	1017	0 = Not used
1.33.12	SP1 Wake Up Level	Varies	Varies	Varies	Varies	1018	0 = Not used
1.33.13	Preset Frequency 1	P1.3	P1.4	Hz	10.0	105	

3.4 Multi-pump (Single Drive) Application

Use the Multi-pump (single drive) application in applications, where 1 drive controls a system that has the maximum of 8 parallel motors, for example, pumps, fans or compressors. By default, Multi-pump (single drive) application is configured for 3 parallel motors.

The drive is connected to 1 of the motors, which becomes the regulating motor. The internal PID controller of the drive controls the speed of the regulating motor and gives control signals by relay outputs to start or stop the auxiliary motors. External contactors (switch) set the auxiliary motors to the mains.

It is possible to control a process variable, the pressure, for example, by the control of the speed of the regulating motor and by the number of motors that operate.

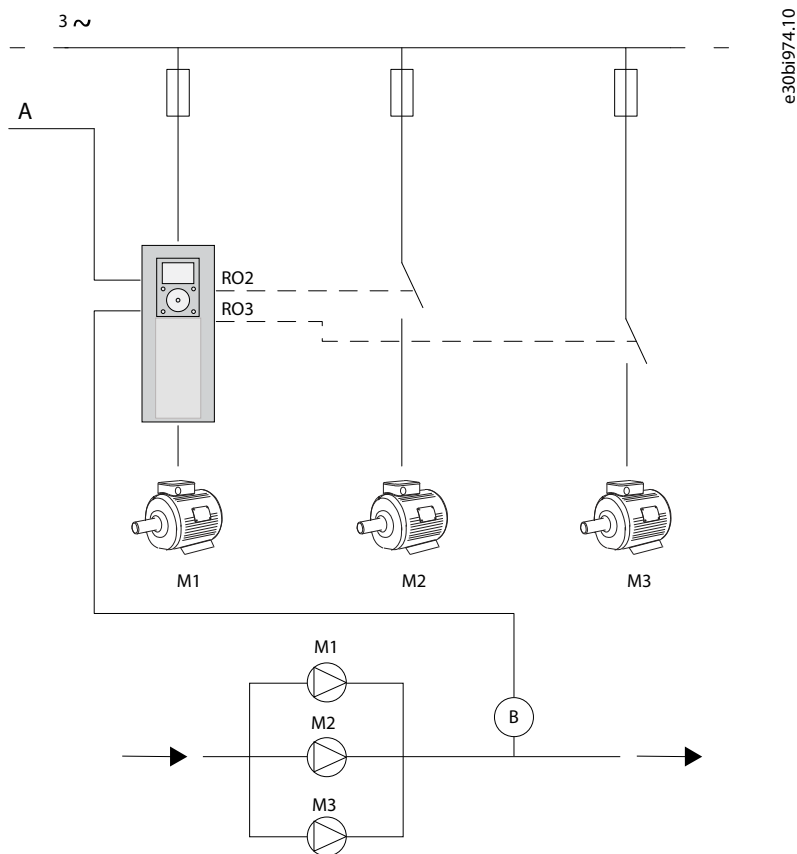


Illustration 10: Multi-pump (Single Drive) Configuration

A	Start / Stop
B	Pressure sensor

Autochange function (change of start order) makes the wear of the motors in the system more equal. Autochange function monitors the running hours and sets the start order of each motor. The motor that has the lowest running hours starts first and the motor that has the highest running hours starts last. It is possible to configure the autochange to start based on the autochange interval time set by the internal real-time clock (an RTC battery needed) of the drive.

It is possible to configure the autochange for all the motors in the system or only the auxiliary motors.

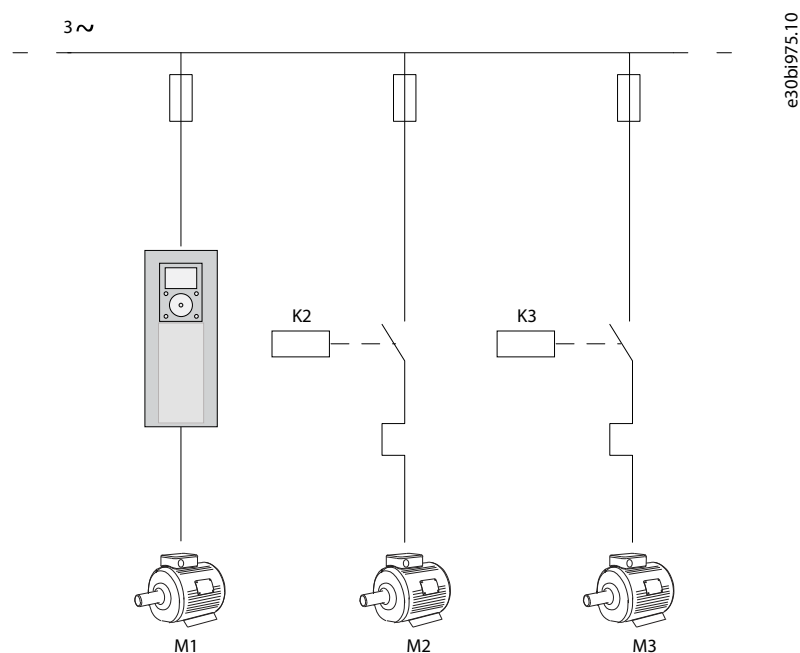


Illustration 11: Control Diagram with Only the Auxiliary Motors Configured to Autochange

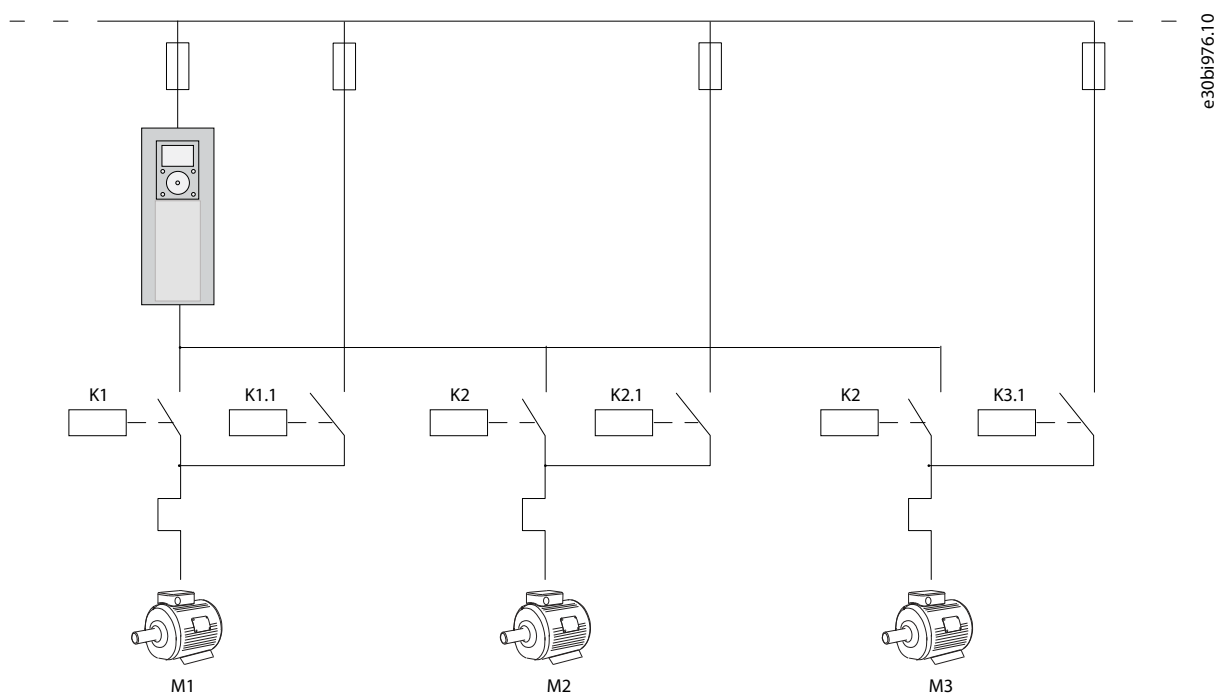


Illustration 12: Control Diagram with All the Motors Configured to Autochange

It is possible to use 2 control places. Make the selection of the control place A or B with DI6. When control place A is active, DI1 gives the start and stop commands, and the PID controller gives the frequency reference. When control place B is active, DI4 gives the start and stop commands, and AI1 gives the frequency reference.

It is possible to configure all the drive outputs freely in all the applications. There is 1 analog output (Output Frequency) and 3 relay outputs (Run, Fault, Ready) available on the basic I/O board.

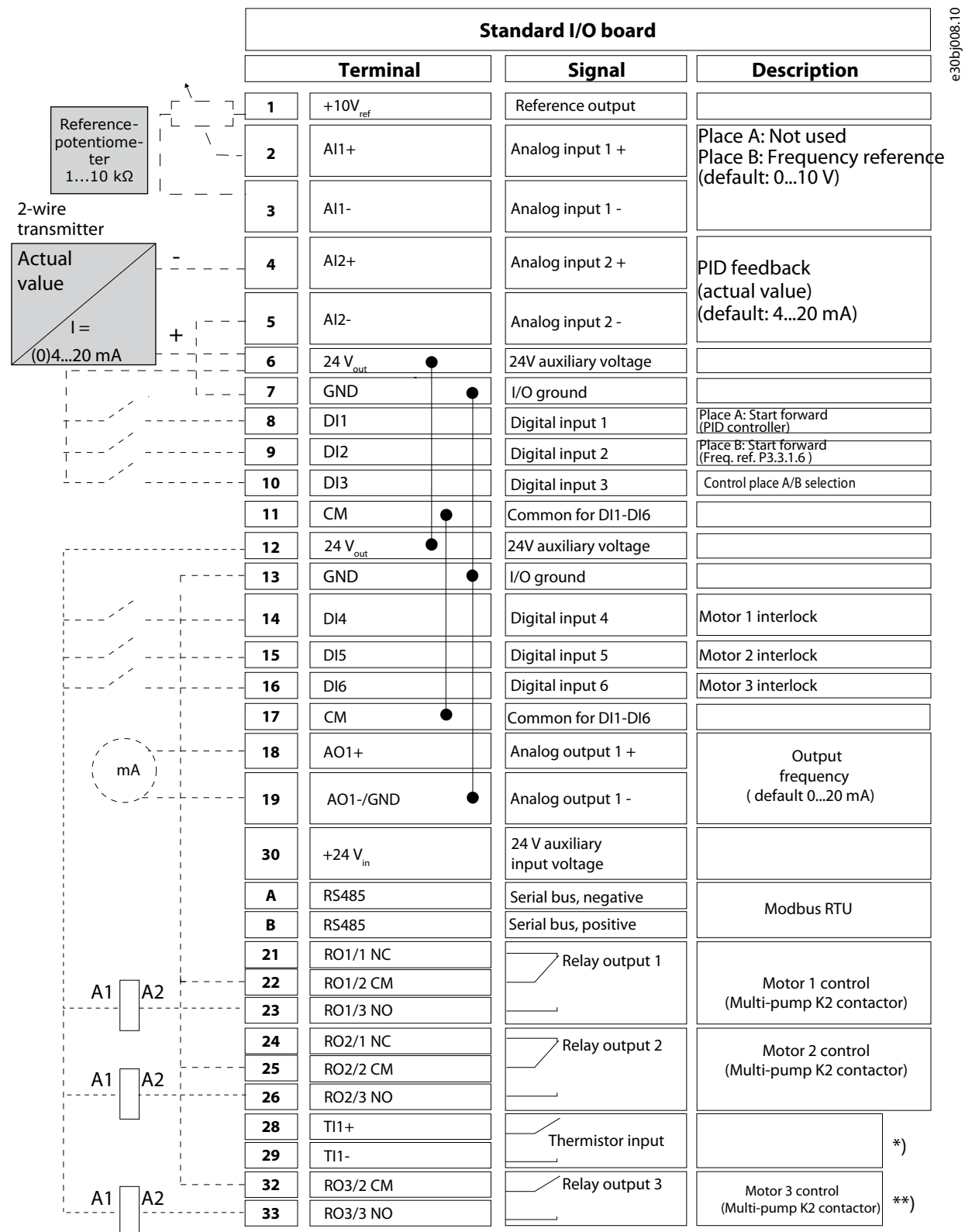


Illustration 13: The Default Control Connections of the Multi-pump (Single Drive) Application

*) Available only in VACON® 100 X. **) For the DIP switch configurations in VACON® 100 X, see the VACON® 100 X Installation manual.

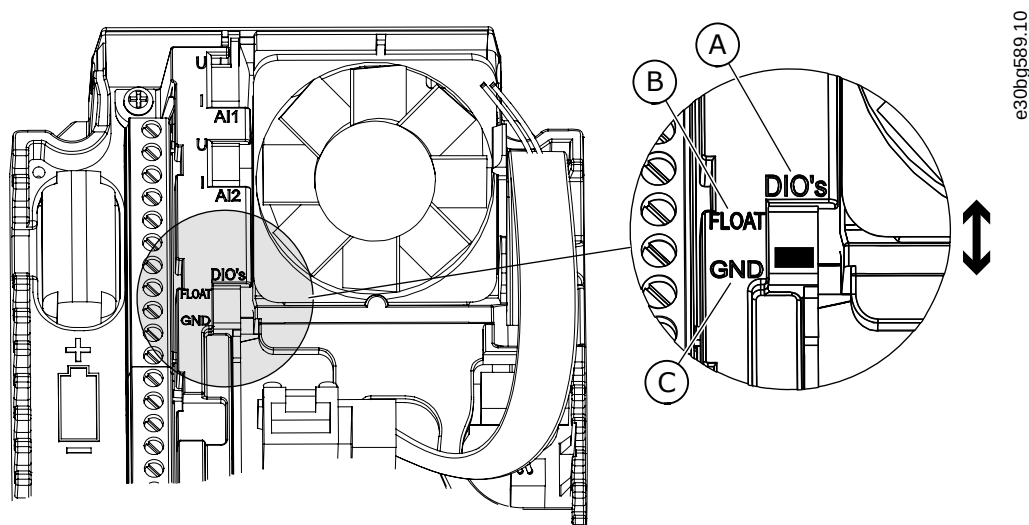


Illustration 14: The DIP Switch

A	Digital input DIP switch	C	Connected to GND (Default)
B	Floating		

See information on the Startup, and Fire mode wizards in [4.2.1 Wizards in Menu 1.1](#).

Table 7: M1 Quick Setup

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.2	Application	0	4		2	212	0 = Standard 1 = HVAC 2 = PID Control 3 = Multi-pump (single drive) 4 = Multi-pump (multidrive)
1.3	Minimum Frequency Reference	0.00	P1.4	Hz	0.0	101	
1.4	Maximum Frequency Reference	P1.3	320.0	Hz	50.0/60.0	102	
1.5	Acceleration Time 1	0.1	3000.0	s	5.0	103	
1.6	Deceleration Time 1	0.1	3000.0	s	5.0	104	
1.7	Motor Current Limit	I _H *0.1	I _S	A	Varies	107	
1.8	Motor Type	0	2		0	650	0 = Induction Motor 1 = Permanent Magnet Motor 2 = Reluctance Motor
1.9	Motor Nominal Voltage	Varies	Varies	V	Varies	110	Find this value U _n on the nameplate of the motor. NOTE! Find out if the motor connection is Delta or Star.
1.10	Motor Nominal Frequency	8.0	320.0	Hz	50/60	111	Find this value f _n on the nameplate of the motor.